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Specialization in
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Identifying Rental Stress Zones in Lisbon

A Spatial Rental Stress Index for Housing Affordability

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Identifying Rental Stress Zones In Lisbon:

A Spatial Rental Stress Index for Housing Affordability

by

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STATEMENT OF INTEGRITY

I hereby declare having conducted this academic work with integrity. I confirm that I have not used plagiarism, any form of undue use of information or falsification of results along the process leading to its elaboration. I further declare that I have fully acknowledged the Rules of Conduct and Code of Honor from the NOVA Information Management School.

Lisbon, 2026/04/27

Diogo Carvalho

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ABSTRACT

This study addresses the growing housing affordability crisis in Lisbon, where existing approaches to measuring rental stress often based on simple income-to-rent ratios, fail to reflect the influence of tourism pressure, urban amenities, and socio-spatial inequalities, limiting their usefulness for urban planning and policy-making. By developing a spatially explicit Rental Stress Index (RSI), this research seeks to provide a more comprehensive measure of rental market pressure. In recent years, rising housing prices driven by tourism, real estate investment, and increased demand have widened the gap between rental costs and household income, intensifying intra-urban inequalities. To overcome the limitations of traditional affordability measures, this research adopts a Design Science Research Methodology (DSRM) to create a composite index integrating socioeconomic, housing, and spatial indicators. The RSI is computed at the parish level using multiple open data sources, including census data, rental prices, and geospatial data on short-term rentals and urban amenities. The results reveal a clear center–periphery pattern, with higher rental stress concentrated in central parishes, where elevated prices, tourism pressure, and urban attractiveness converge. Peripheral areas show comparatively lower stress levels. The findings demonstrate that rental stress is multidimensional, emerging from the interaction between housing market dynamics, socioeconomic vulnerability, and territorial characteristics. An interactive Power BI dashboard was developed to support the visualization and interpretation of results, enabling more informed decision-making. This study contributes a replicable, data-driven framework to support housing policy and promote more equitable urban development.

KEYWORDS

Rental Stress; Housing Affordability; Urban Analytics; Composite Indicators; Spatial Decision Support; Short-Term Rentals

Sustainable Development Goals (SDG):



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LIST OF ABBREVIATIONS AND ACRONYMS

BI	Business Intelligence
DSRM	Design Science Research Methodology
GIS	Geographical Information Systems
INE	Instituto Nacional de Estadística
KMO	Kaiser-Meyer-Olkin test
LISA	Local Indicators of Spatial Association
PCA	Principal Component Analysis
POI	Point of Interest
RSI	Rental Stress Index
RQ	Research Question
STR	Short-Term Rental



1. INTRODUCTION

Lisbon has become a paradigmatic case of the housing affordability crisis that affects many European cities. Over the past decade, the city has experienced an unprecedented escalation in rental and property prices, widening the gap between housing costs and household income levels. As Lorga et al. (2022) point out, the commoditization of real estate investment, combined with rapid tourism growth and foreign demand, has severely eroded housing affordability in the Portuguese capital. This dynamic has contributed to the displacement of low- and middle-income residents from central neighborhoods to peripheral areas, reshaping the city's social and spatial structure. The authors emphasize that these transformations are not isolated events but the product of economic, political, and social interactions that reflect broader urban and financial dynamics.

The housing affordability crisis in Lisbon exemplifies a global challenge, in which housing has increasingly shifted from a social good to a financial asset. Following the financial crisis of 2008, many European housing markets, including Portugal's, became highly exposed to speculative investment and policy deregulation (Cocola-Gant & Gago, 2021; Tulumello & Allegretti, 2021). In Lisbon, this transformation has been particularly intense due to the city's appeal to international investors and tourists, exacerbating *rental stress*. As identified by Lee et al. (2022) and Mineshima et al. (2021), the structural nature of the affordability problem calls for innovative analytical tools capable of identifying spatial patterns of housing pressure and guiding more equitable public policies.

Despite the increasing visibility of the problem, tools for identifying and visualizing *rental stress* remain limited, often relying on static, non-spatial, and outdated data. Real estate platforms typically neglect the geographic dimension of housing markets, failing to account for how location influences affordability (Schernthanner et al., 2016). Similarly, conventional data sources, such as census statistics, fail to capture the fast-paced urban transformations observed in Lisbon and other major cities (Boeing & Waddell, 2016). This lack of spatially aware, dynamic analytical frameworks hinders researchers' and policymakers' capacity to monitor and respond effectively to local housing stress.

To address this gap, the present research aims to answer the following question: How can we identify urban areas in Lisbon that are at risk of losing rental housing affordability? The study's main objective is to develop a Rental Stress Index (RSI) that quantifies and visualizes the spatial distribution of rental stress across Lisbon. More specifically, it seeks to construct a composite index that integrates spatial and socioeconomic indicators, apply it at the neighborhood level to uncover intra-urban disparities, and implement an interactive Power BI dashboard that supports evidence-based decision-making in housing and urban policy.

This research adopts the Design Science Research Methodology (DSRM) (Peppers et al., 2007a), an approach particularly suited to the creation of innovative artifacts that solve real-world problems. The methodology follows five iterative phases, including problem identification,



design, development, demonstration, and evaluation, which structure the development of both the Rental Stress Index and the interactive dashboard. The study integrates spatial and socioeconomic datasets from multiple open sources, including OpenStreetMap, Geodados Lisboa, InsideAirbnb, Lisboa Aberta, INE, and Eurostat, and aggregates them at the neighborhood level using Geographic Information Systems (GIS), then visualizes them in Power BI. The resulting artifact combines analytical precision with interactive usability, supporting data-driven urban management.

The expected contribution of this research is twofold. First, it proposes a replicable methodological framework that integrates spatial and socioeconomic data to analyze rental stress dynamically. Second, it provides a practical decision-support tool that enhances transparency, informs public policy, and promotes urban equity. By visualizing patterns of rental stress in Lisbon, this research not only deepens the understanding of the city's housing dynamics but also contributes to the broader debate on data-informed and sustainable urban governance (Tanji et al., 2024). Ultimately, it aims to support more just and evidence-based housing policies, fostering a fairer and more inclusive city.

Accordingly, this research is guided by the following research questions:

RQ1: How can a Rental Stress Index (RSI) be developed, using socioeconomic, housing, and territorial indicators, to measure and compare rental stress across Lisbon's parishes?

RQ2: What spatial patterns of rental stress emerge in Lisbon when applying the RSI at the parish level, and what intra-urban inequalities do these patterns reveal?

RQ3: To what extent can an interactive Power BI dashboard support the monitoring and interpretation of the RSI, contributing to evidence-based decision-making and the prioritization of housing policies in Lisbon?

This study offers both methodological and practical contributions. Methodologically, it proposes a replicable framework for constructing a spatially explicit Rental Stress Index by integrating housing, socioeconomic, and territorial indicators into a composite model. Practically, it delivers an interactive dashboard that supports the monitoring and interpretation of rental stress, helping policymakers and urban planners identify priority areas for intervention. By combining urban analytics, spatial data, and business intelligence tools, the study contributes to data-driven housing governance and evidence-based urban planning.

This thesis is structured as follows. Chapter 2 reviews the literature on housing affordability and rental stress. Chapter 3 presents the methodology, including data sources, RSI construction, and dashboard development. Chapter 4 presents and discusses the empirical results. Finally, Chapter 5 concludes the study and outlines directions for future research.



2. LITERATURE REVIEW

2.1 UNDERSTANDING HOUSING AFFORDABILITY AND RENTAL STRESS

The literature on housing affordability and rental stress highlights the growing complexity of urban dynamics and inequalities in access to housing. Recent studies emphasize the impact of financialization, tourism, and urban policies on the intensification of housing pressure, particularly in cities like Lisbon (Cocola-Gant & Gago, 2021; Lorga et al., 2022). Within this context, emerging approaches integrate socioeconomic, spatial, and environmental data to more accurately measure housing accessibility. This review examines the key concepts, variables, and methodologies underpinning the development of the Rental Stress Index (RSI) and contributes to a spatially informed and integrated understanding of the housing crisis.

2.2 CORE VARIABLES RELATED TO RENTAL STRESS

Designing a spatially explicit Rental Stress Index (RSI) requires a multidimensional approach that captures the complexity of housing affordability. Rental stress emerges from the interaction of economic, sociodemographic, spatial, environmental, and institutional factors, which together reflect structural inequalities and territorial dynamics that shape urban housing systems.

Social attributes shape rent-paying stress in multiple, mutually reinforcing ways. Limited social support networks reduce households' ability to cope with unexpected financial burdens, intensifying rental pressure among low-income tenants (Ekta & Vardhan, 2025). Socio-demographic factors, such as age, household size, occupation, and income stability, further condition affordability, with vulnerable groups experiencing disproportionate strain as a larger share of income is absorbed by rent (Amin, 2025).

Market-related factors also play a crucial role in determining housing stress. The characteristics of dwellings, their location, and the overall quality of the built environment strongly influence rent levels. Hedonic models developed in Santiago (Vergara-Perucich, 2023) and Belém (Freitas et al., 2024) show that structural and locational attributes explain much of the variation in rent prices.

Spatial indicators situate rental stress within its geographic and functional context, emphasizing how tourism, gentrification, and accessibility shape urban housing markets. In tourism-driven cities like Lisbon, the concentration of short-term rentals (STRs) is directly linked to rising rents and the displacement of long-term residents (Lestegás et al., 2019a; Orozco Martinez & Gil-Alonso, 2024). Digital rental platforms such as Airbnb restructure residential markets by concentrating profits among a few owners and displacing traditional tenants (Robertson et al., 2022), while Song et al. (2024) find that tourism development raises



housing prices through investment and income effects. Access to transport and essential services also affects rental affordability. Yang et al. (2020) and Kou et al. (2024) demonstrate that proximity to schools, hospitals, and transit hubs significantly increases rents, particularly in peripheral neighborhoods. Green spaces also play a significant role in shaping rental stress, as their proximity is capitalized into housing prices in most urban contexts. Empirical evidence shows that access to parks, urban forests, and other green amenities increases residential property values by improving environmental quality, offering recreational opportunities, and mitigating urban heat (Cheng et al., 2024; Łaskiewicz et al., 2022).

2.3 CONCEPTS AND MEASURES OF HOUSING AFFORDABILITY AND RENTAL STRESS

Operationalizing housing affordability reveals deep methodological divides that affect both empirical measurement and policy design. The dominant paradigm, ratio-based measures such as the 30% expenditure-to-income rule, offers simplicity but suffers major conceptual weaknesses, including the omission of household composition, non-housing costs, financing structures, and spatial price variation (Dancygier & Wiedemann, 2025; Lorga et al., 2022).

Residual-income approaches improve on this by accounting for non-housing consumption and family structure, thereby better capturing lived housing stress across income groups and urban contexts (Biljanovska, 2023). Yet, these methods are data-intensive and rely on assumptions about minimum consumption baskets, which limit their policy usability (Rangel et al., 2017).

Emerging multidimensional or composite indices attempt to integrate multiple indicators, including income ratios, payment burdens, distributional metrics, and housing quality, into unified frameworks (Dancygier & Wiedemann, 2025; Gonçalves et al., 2023). These indices enhance cross-city comparison and predictive accuracy but raise concerns about weighting subjectivity, methodological transparency, and limited spatial resolution, which is essential to capture neighborhood-level dynamics such as gentrification and STR concentration (Lee & Kim, 2023)

This methodological landscape reveals a central trade-off: simpler measures are policy-friendly but conceptually thin, while complex multidimensional approaches are richer but less operational. The challenge for rental stress measurement lies in achieving both analytical rigor and spatial specificity.



2.4 METHODOLOGICAL APPROACHES IN RENTAL STRESS MEASUREMENT

Four major methodological approaches dominate current research on rental stress. Composite indices synthesize multiple variables by normalizing, weighting, and aggregating them, such as using z-scores or principal component analysis (PCA). They provide holistic, comparable metrics but introduce subjectivity and risk, obscuring inequality within aggregated scores (Dancygier & Wiedemann, 2025; Gonçalves et al., 2023).

Hedonic pricing models decompose housing prices into implicit values for characteristics and location, isolating factors such as STR density or accessibility. Quasi-experimental variants that exploit Lisbon's STR regulations offer credible causal identification (Gonçalves et al., 2023). However, these models require rich data and measure price effects rather than affordability directly.

GIS-based spatial analysis captures the geography of housing stress. Techniques such as Moran's I and Local Indicators of Spatial Association (LISA) detect clustering, while spatial regressions account for spatial dependence (Barreca et al., 2018). These methods are vital for neighborhood-level insights but require spatial expertise and detailed geodata (Goodman et al., 2018).

Dashboards and dynamic visualizations translate analysis into policy tools. Integrating GIS and Business Intelligence (BI) allows real-time monitoring and scenario testing (Wickramasuriya et al., 2014). However, adoption remains limited, especially in Southern Europe, where institutional capacity lags.

The methodological frontier lies in integrating these approaches by combining spatially explicit composite indices, causal validation, and BI dashboards, thus bridging the gap between academic sophistication and policy utility.

2.5 RESEARCH GAPS AND THE NEED FOR THE RSI

Despite growing scholarship, three critical gaps persist in measuring rental stress and informing policy, particularly for Lisbon:

Gap 1: The lack of spatialized neighborhood-level metrics. Existing affordability measures operate mostly at metropolitan or national scales, masking intra-urban disparities. Lisbon lacks a tool for tracking rental stress across neighborhoods, despite abundant open data (Antunes & Ferreira, 2021; Mendes, 2021).

Gap 2: The limited integration of open-source spatial and socioeconomic data. While multidimensional indices exist, few synthesize diverse open datasets such as STR listings, cadastral data, census microdata, or transport networks into unified analytic frameworks (Riedler & Lang, 2022).



Gap 3: The scarcity of decision-support systems combining GIS with BI. Few examples integrate spatial analysis and dashboards for monitoring or policy evaluation.

The proposed Rental Stress Index (RSI) addresses these gaps by integrating residual-income and ratio measures, STR exposure, and displacement indicator into a spatially explicit composite index. By combining composite methodologies, spatial econometrics, and quasi-experimental validation within an interactive Power BI dashboard, the RSI offers a policy-oriented, neighborhood-level tool for housing governance in Lisbon. It also provides a transferable framework for other European cities facing similar pressures (Colomb & Moreira De Souza, 2021; Gasbarri et al., 2024; Koch, 2023; Lestegás et al., 2019b; Marque et al., 2023)



3. METHODOLOGY

3.1 DESIGN SCIENCE RESEARCH

The Design Science Research Methodology (DSRM) is particularly suitable for this study because it provides a structured process for developing innovative artifacts that address real-world problems, such as the need to identify and monitor rental stress in Lisbon. DSRM guides the creation of the Rental Stress Index (RSI) and the Power BI dashboard through sequential stages, from problem identification and objective definition to the design, development, demonstration, and evaluation of the artifact.

This approach ensures methodological rigor, supports the integration of diverse spatial and socioeconomic data, and guarantees that the RSI emerges as a useful, transparent, and replicable decision-support tool for urban policy and housing governance (Peffer et al., 2007).

As illustrated in Figure 1, the process follows an iterative sequence. It begins with problem identification, establishing housing affordability and tourism pressure as core challenges. The objectives phase focuses on developing the RSI and integrating relevant data sources. The design and development stage involves constructing the artifact, including applying factor analysis and implementing it in Power BI. The demonstration phase applies the RSI to Lisbon's parishes, enabling spatial exploration through the dashboard. The evaluation stage assesses both the validity of the results and the analytical usefulness of the tool, while the final communication phase ensures the dissemination of findings and their policy relevance.

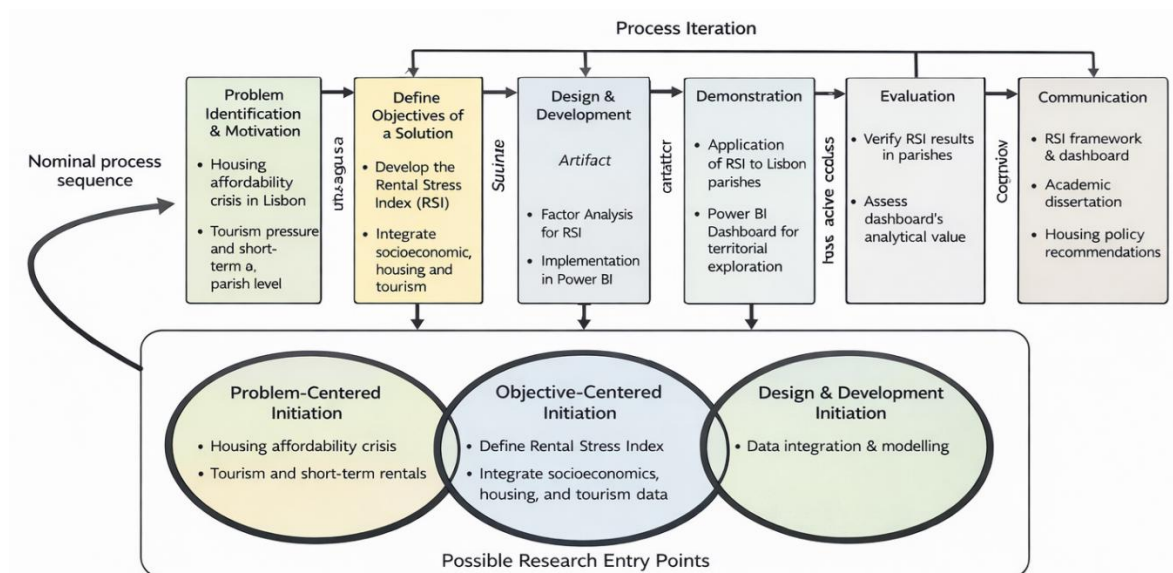


Figure 1 - Methodological Framework. Adapted from Peffer (2007)



3.2 STUDY AREA

Although several indicators are computed for all parishes in Lisbon to provide a citywide contextual overview (Figure 2), the analytical development of the Rental Stress Index (RSI) is intentionally restricted to a set of central parishes. Lisbon is adopted as the study area due to its pronounced housing affordability pressures, strongly shaped by tourism growth, short-term rentals, and intense real estate dynamics over the past decade.

The focus on central parishes reflects their role as the structural core of rental stress in the city. These areas concentrate higher densities of short-term rental units, a more competitive long-term rental market, and stronger tourism activity, making them the locations where the relationship between tourism and housing affordability is most persistent and pronounced. Restricting the RSI to these parishes avoids diluting the index with peripheral areas whose housing and tourism dynamics differ substantially.

This spatial delimitation is further supported by empirical evidence showing that Lisbon's central parishes largely coincide with the city's main north–south subway axis, a corridor consistently associated with higher residential property values. In their study, Trindade Neves et al. (2024) identify proximity to the subway as one of the most influential determinants of housing prices in Lisbon, reinforcing the relevance of focusing the RSI on these central, highly accessible areas where market pressures converge most clearly. In Figure 1, the legend distinguishes parishes included in the RSI analytical scope from those excluded: parishes highlighted in red (“Yes”) correspond to the central areas where the Rental Stress Index is computed. In contrast, parishes shown in green (“No”) represent the remaining areas of Lisbon used only to provide a broader citywide contextual reference.

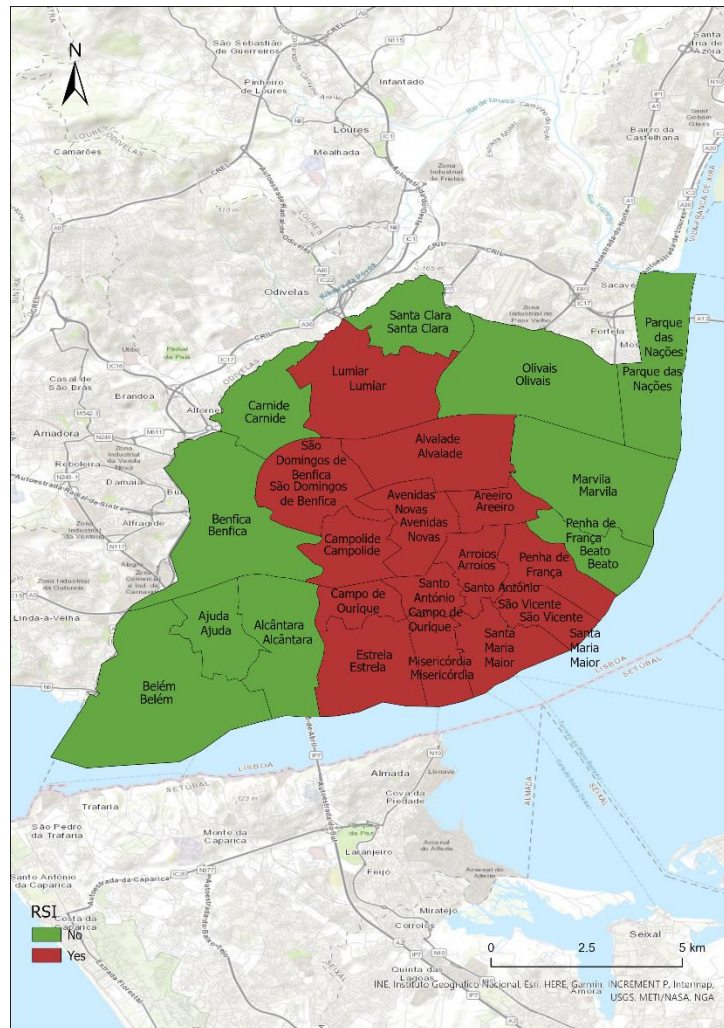


Figure 2 - Study Area

3.3 DATA SOURCES

In the development of the analytical model underlying the Rental Stress Index (RSI), the main business analytical processes related to the characterization of the rental market and the socioeconomic conditions of Lisbon's resident population were identified. These processes include the analysis of rent evolution over time, the socioeconomic characterization of parishes, and the assessment of the urban context through the spatial distribution of points of interest (POIs), as shown in Table 1.



Table 1 - Variables, Data Sources, and Supporting Literature

Variable	Description	Data Source ¹	Supporting Literature	Spatio-temporal
Total Classic Dwellings for Rent	Total number of classic dwellings used for rental purposes, regardless of the type of rental contract.	INE	(Alves et al., 2023)	Parish/Year (2021)
Fixed-term Rental Contracts	Number of rental contracts with a defined duration and a pre-established end date.	INE	(Alves et al., 2023)	Parish/Year (2021)
Open-ended Rental Contracts	Number of rental contracts with no fixed end date, remaining valid until terminated by one of the parties.	INE	(Alves et al., 2023)	Parish/Year (2021)
Total Classic Dwellings	Total number of classic dwellings, including occupied, vacant, and secondary residences, excluding collective housing units.	INE	(Alves et al., 2023)	Parish/Year (2021)
Unemployed Population	Number of individuals without employment who are actively seeking work and available to work, according to census definitions.	INE	(Trindade-Neves et al. 2024)	Parish/Year (2021)
Total Population	Total number of usual residents in a given geographical area.	INE	(Amin 2025)	Parish/Year (2021)
Population Under 15	Number of residents aged under 15 years, representing the young population.	INE	(Amin 2025)	Parish/Year (2021)
Working-age Population	Population of working age, typically defined as individuals aged between 15 and 64 years, including both employed and unemployed persons.	INE	(Amin 2025)	Parish/Year (2021)
Population Aged 65 and Over	Number of residents aged 65 years and over, representing the elderly population.	INE	(Amin 2025)	Parish/Year (2021)
AirBNB	Geographical Distribution of Airbnb's	Inside AirBnB	(Orozco Martinez & Gil-Alonso, 2024)	Parish/Year (2026)
POI_Mobility	Subway, Fluvial, and Train stations	Geodados Lisboa	(Trindade-Neves et al. 2024)	Parish/Year (2026)
POI_Education	School Facilities (public and private schools, basic, secondary, and professional schools)	Geodados Lisboa	(Trindade-Neves et al., 2024)	Parish/Year (2026)
POI_Commerce	Markets and Malls	Geodados Lisboa	(Trindade-Neves et al., 2024)	Parish/Year (2026)
POI_Safety and Security	Public safety and emergency services, including police services, fire departments, and civil protection authorities	Geodados Lisboa	(Trindade-Neves et al., 2024)	Parish/Year (2026)

¹ Source: Geodados Lisboa: <https://geodados-cml.hub.arcgis.com/>; INE: <https://www.ine.pt>; Idealista: <https://www.idealista.pt/>; Inside Airbnb: <https://insideairbnb.com/>



Variable	Description	Data Source ¹	Supporting Literature	Spatio-temporal
POI_Culture	Refers to points of interest related to cultural infrastructure and activities, including auditoriums and amphitheaters, libraries, archives and documentation centers, cinemas, museums, theatres, cultural associations, art galleries, cultural centers, artistic residencies, regional cultural houses, and cultural foundations.	Geodados Lisboa	Trindade-Neves & Castro (2024)	Parish/Year (2026)
Rental Price per Square Meter	Rental price per square meter (€/m ²)	Idealista	(St-Hilaire et al., 2024)	Parish/Monthly

3.4 DESCRIPTIVE STATISTICS

Descriptive statistics represent a fundamental step in the preliminary exploration of the data used to develop the Rental Stress Index (RSI). This analysis provides an initial understanding of the structure, scale, and variability of the variables, allowing for the identification of general patterns, dispersion levels, and potential extreme values. Such an assessment is essential to ensure the analytical suitability of the dataset prior to normalization and multivariate modeling.

Table 2 summarizes the descriptive statistics for all variables included in the analysis, reporting the number of observations (Obs), mean (Avg), minimum (Min) and maximum (Max) values, standard deviation (Std), and data type. All variables are computed at the parish level, in accordance with the grain defined in the dimensional model, ensuring spatial consistency across the different data sources integrated into the RSI framework.

It is important to highlight that no null or missing values were identified in any of the variables considered. This reflects a prior process of data validation, cleaning, and harmonization across heterogeneous sources. The absence of missing data eliminates the need for imputation procedures, reduces the risk of statistical bias, and enhances the robustness and reliability of subsequent normalization and factor analysis steps.

The variables related to Points of Interest (POIs) exhibit substantial spatial heterogeneity across parishes, particularly in the categories of culture, leisure, and education, as indicated by relatively high standard deviations. This variability reflects the functional differentiation of Lisbon's urban space and supports the inclusion of POIs as contextual indicators of urban attractiveness and territorial pressure.

Regarding housing market variables, considerable dispersion is observed in the number of classic dwellings allocated to rental use and in the distribution of contract types (fixed-term versus open-ended contracts). These differences reveal structural inequalities in rental market stability across parishes, which are directly relevant to the assessment of rental stress, as more unstable rental structures tend to increase residential vulnerability.



Table 2 - Summary Statistics (Parish Level, N=24)

Feature	Avg	Max	Min	Std	Data Type
POI_Mobility	3.50	9.00	0.00	2.52	Numerical
POI_Education	9.38	19.00	3.00	4.57	Numerical
AirBNB	724.17	3493.00	35.00	882.97	Numerical
POI_Commerce	4.13	12.00	0.00	3.27	Numerical
POI_Culture	29.25	89.00	2.00	23.76	Numerical
POI_Leisure	15.75	33.00	4.00	7.82	Numerical
POI_SafetyAndSecurity	4.96	17.00	0.00	4.07	Numerical
Total Classic Dwellings for Renting	4262.96	7674.00	2294.00	1480.65	Numerical
Fixed-term Rental Contracts	2234.33	4643.00	984.00	862.08	Numerical
Open-ended term Rental Contracts	2028.63	5006.00	934.00	864.98	Numerical
Total Classic Dwellings	10085.17	19547.00	4545.00	4174.75	Numerical
Total Unemployed Population	921.29	2111.00	370.00	410.46	Numerical
Total Population	22741.50	46334.00	9658.00	9775.35	Numerical
Population aged 15 and Under	2967.50	6790.00	875.00	1418.57	Numerical
Working-age Population	14450.33	29855.00	6472.00	6157.55	Numerical
Population Aged 65 and Over	5323.67	10475.00	2027.00	2481.12	Numerical
Rental Price (€/m ² /month)	18.21	29.20	11.00	4.65	Numerical

The socioeconomic and demographic variables also display pronounced variability, particularly in total population, working-age population, and elderly population, highlighting distinct demographic profiles and differing capacities to absorb rental market pressure. Although unemployment levels are lower on average, their spatial variation remains an important indicator of economic vulnerability at the local level.

The rental price per square meter (€/m²), derived from monthly observations, shows a wide range between minimum and maximum values, confirming the strong spatial segmentation of Lisbon's rental market. This pattern reinforces the need for a composite and spatially explicit approach, such as the RSI, to capture the multidimensional nature of rental stress across neighborhoods.

3.5 DEVELOPING THE RSI

The Rental Stress Index (RSI) is a composite indicator designed to measure relative levels of stress in the rental housing market at the parish level. Conceptually, the RSI adopts a multidimensional approach to rental stress, moving beyond traditional affordability measures



based solely on the income–rent relationship. Instead, it integrates housing market characteristics, socioeconomic conditions, and territorial pressures, in line with the literature on rental stress and rental vulnerability (Troy & Martin, 2017).

The selection of variables reflects four main dimensions: housing market conditions (e.g. price per square metre, rental share), socioeconomic context (e.g. unemployment rate, ageing index), rental dynamics (e.g., intensity of short-term rentals), and urban pressures associated with the presence of amenities and services (points of interest related to education, commerce, culture, mobility and security). The core Rental Stress Index is constructed using housing, socioeconomic, and labor market indicators, ensuring conceptual validity. Points of Interest are analyzed separately as contextual urban attractiveness factors, providing complementary insights for policy interpretation and dashboard visualization.

This approach is consistent with established methodologies for constructing territorial composite indicators in urban and regional studies (Jardim et al., 2022).

After the creation of derived variables, all indicators were standardized using z-score normalization to ensure comparability across different measurement scales.

The RSI was estimated using Factor Analysis, extracting a single latent factor interpreted as the overall level of rental stress. This method is widely used in the development of composite territorial indicators due to its robustness and interpretability (Jardim et al., 2022).

The resulting factor scores were rescaled to a 0–100 range, facilitating interpretation and spatial comparison between parishes. Higher RSI values indicate higher levels of rental stress.

However, as an aggregated indicator, the RSI does not directly reveal which factors explain higher or lower levels of housing stress in each territory. Therefore, a complementary analysis was conducted to support the territorial interpretation of the index.

To assess the suitability of the data for Factor Analysis, the Kaiser-Meyer-Olkin (KMO) measure and Bartlett's Test of Sphericity were conducted. The KMO value of 0.614 indicates adequate sampling adequacy, while Bartlett's test was statistically significant ($p < 0.05$), confirming that the correlations between variables are sufficiently strong for factor extraction.

The factor extraction yielded a single dominant component, accounting for 72.07% of the total variance, supporting the use of a unidimensional latent construct to represent rental stress.

For each indicator, deviations from the citywide average were calculated and subsequently normalized to ensure comparability across variables with different units. This made it possible to identify, for each parish, the factors most associated with relatively higher or lower housing stress levels.



This analytical procedure was implemented in Power BI, enabling interactive exploration of the results. Importantly, this step does not modify the construction of the RSI but rather supports the territorial interpretation of the index results.

To ensure transparency and reproducibility, all data sources (when permitted) and the Python scripts used for data processing, transformation, and analysis are available in a public GitHub repository: <https://github.com/DiogoFcarvalho/RSI-Dashboard>

3.6 DEVELOPING THE DASHBOARD

The development of the dashboard followed a structured, needs-driven approach, starting from the identification of the key analytical and decision-making challenges associated with the rental housing market. Rather than focusing on isolated indicators, the process began with the definition of business needs, understood as the core questions that stakeholders, planners, and policymakers need to answer to better understand housing dynamics and territorial inequalities.

These business needs were formulated to capture the complexity of rental market pressure, combining housing supply characteristics, socioeconomic conditions, and the influence of tourist accommodation. Defining the business needs at an early stage was essential to ensure that the dashboard would not merely present data but instead function as a decision-support tool capable of addressing real policy and planning concerns.

Once the business needs were clearly established, they were translated into analytical questions and measurable indicators that could be operationalized using available data sources. This translation process followed the Kimball approach to dimensional modeling, in which business requirements drive the design of the data model and the definition of metrics (Kimball, 2013). In this context, the identification of relevant variables, the selection of appropriate spatial units of analysis (parish level), and the standardization of indicators across territories were guided by the principle of aligning analytical outputs with decision-making needs.

3.6.1 BUSINESS NEEDS – RENTAL MARKET PRESSURE AND HOUSING VULNERABILITY

The purpose of this dashboard is to support a comprehensive analysis of rental market pressure and housing vulnerability at the parish level by integrating housing, socioeconomic, and tourism-related indicators. By combining multiple dimensions of the housing system, the dashboard enables a nuanced understanding of how rental supply, household income, contract structures, and tourist accommodation interact spatially and contribute to unequal access to housing.



The dashboard is designed to identify territories experiencing higher levels of housing vulnerability, where the availability of long-term rental housing is limited, household purchasing power is relatively low, and competition from short-term rental activities is more intense. Through a spatially explicit approach, it allows users to detect parishes where structural imbalances in the housing marketplace increasing pressure on permanent residents, supporting evidence-based territorial diagnosis.

A central component of the analysis concerns the structure of the rental market, particularly the relationship between total housing stock and dwellings allocated to the rental sector, as well as the prevalence of short-term versus open-ended rental contracts. This perspective helps to assess market stability and identify areas where rental conditions are more precarious or volatile, potentially increasing the risk of residential displacement.

The dashboard also addresses housing affordability and socioeconomic context by analyzing household income levels in relation to the availability and characteristics of rental housing. This enables the identification of parishes where lower income levels coincide with constrained rental supply, highlighting areas with heightened affordability stress and reduced capacity for households to absorb rising housing costs.

Another key analytical dimension is the impact of tourist accommodation on residential housing markets. The growing concentration of short-term tourist units introduces direct competition with long-term rental housing, influencing both supply and pricing dynamics. By examining the spatial relationship between tourist accommodation density and classic rental availability, the dashboard enables identification of areas where tourism intensifies pressure on the residential housing market.

By integrating these housing and socioeconomic dimensions in an analytical framework, the dashboard provides a decision-support tool for urban and housing planning. It facilitates prioritizing territories for policy intervention, monitoring spatial inequalities in access to housing, and developing data-driven strategies to promote more balanced and inclusive housing systems.

3.6.2 KEY ANALYTICAL QUESTIONS

To operationalize the identified business needs, a set of key analytical questions was defined across different dimensions, as presented in Table 3.

Table 3 - Key Analytical Questions

Analytical Dimension	Key Analytical Questions
Rental Supply and Housing Pressure	– Which parishes have the lowest number of classic dwellings available for rent? – What is the relationship between the total number of classic dwellings and those allocated to the rental market?



Analytical Dimension	Key Analytical Questions
	– Which parishes show a higher proportion of short-term rental contracts compared to open-ended contracts?
Income and Affordability	– In which parishes is the average household income lowest relative to the available rental housing supply? – Is there a correlation between lower income levels and higher rental market pressure? – Which parishes present the highest potential housing affordability stress?
Tourist Accommodation and Market Competition	– Which parishes have the highest density of tourist accommodation units (e.g., Airbnb)? – In which areas does a high presence of tourist accommodation coincide with a reduced supply of classic rental housing? – Is there a relationship between tourism intensity and the decline of long-term rental contracts?
Territorial Prioritization and Policy Support	– Which parishes exhibit the highest overall levels of rental market pressure? – Which housing, socioeconomic, and tourism-related dimensions contribute most to market pressure in each parish? – Which parishes should be prioritized for housing and urban policy intervention?

3.7 DATA TRANSFORMATION

Data preparation constituted a crucial stage in the development of the Rental Stress Index (RSI), ensuring the consistency, comparability, and integration of information from multiple statistical, spatial, and market data sources. For this purpose, Power Query, integrated within Power BI, was used as the main tool, given its suitability for data cleaning, harmonization, aggregation, and the creation of analytical indicators.

In an initial phase, data cleaning and validation procedures were carried out, including the identification of missing or inconsistent values, the standardization of formats, and the normalization of territorial designations. These steps were essential to guarantee compatibility across datasets originating from different sources.

Subsequently, units of measurement were harmonized. Rental values were initially expressed in euros per square meter (€/m²) and were converted into absolute monthly monetary values (€). Based on this transformation, an estimated annual rent value was calculated by multiplying the monthly rent by twelve, ensuring temporal consistency with the annual income indicators used in the analysis.

Spatial aggregation procedures were then applied, consolidating data at the freguesia level, which was adopted as the territorial unit of analysis. This process was applied to both census data and georeferenced datasets, particularly in the aggregation of points of interest (POIs), allowing for a detailed characterization of the urban and functional context of each territory.

Within the creation of derived variables, a set of analytical indicators was developed to operationalize the concepts of housing stress and socioeconomic vulnerability. Among these,



particular emphasis is placed on the ratio between estimated annual rent and annual income, calculated as the quotient between annual rental costs and average annual disposable income. This indicator provides an approximation of the financial burden faced by households in relation to housing costs, with higher values indicating greater pressure of housing expenses on available income.

Additional derived indicators were also computed in order to capture complementary dimensions of housing stress and to contextualize rental affordability within broader structural, market, and demographic dynamics:

- 1) Share of traditional rental housing within the total housing stock, calculated as the ratio between the number of classic dwellings intended for rental and the total number of classic dwellings. This indicator is relevant as it reflects the availability of long-term rental housing in each territory. Lower shares may indicate constrained supply, which can intensify competition in the rental market and contribute to higher rental prices and increased housing stress.
- 2) Structure of rental contracts, expressed through the proportion of fixed-term contracts relative to open-ended contracts. This variable is used as a proxy for rental market stability, as a higher prevalence of fixed-term contracts is commonly associated with greater insecurity for tenants, reduced residential stability, and increased vulnerability to rent increases or displacement.
- 3) Indicators of tourism pressure, derived from the relationship between the number of tourist accommodation units and the number of dwellings allocated to traditional rental housing. This indicator is central to the study, as tourism-driven demand can directly compete with residential rental markets, reducing housing availability for permanent residents and exerting upward pressure on rents, particularly in central and highly attractive urban areas.
- 4) Demographic rates and indices, such as the unemployment rate and the aging index, are derived from the relationships between different population groups. These indicators are essential for identifying socioeconomic vulnerability, as higher unemployment levels are associated with reduced household income and a greater risk of housing cost overburden. At the same time, aging populations may face increased exposure to housing stress due to fixed or declining incomes and specific housing needs.

3.8 DATA MODEL

Based on these processes, the grain of the dimensional model was defined as the parish at a specific point in time, ensuring spatial and temporal consistency across the different integrated data sources.

The model is built upon two dimensions:



- Parish Dimension, which represents the territorial unit of analysis and enables the integration of statistical, geographic, and market data at the intra-urban level;
- Time Dimension, which supports temporal analysis and is particularly relevant for examining rental price evolution.

From these dimensions, three fact tables were designed, each corresponding to a distinct analytical process:

- Fact POIs, which store metrics related to the presence and density of points of interest by parish, allowing for the characterization of the urban context and accessibility to services and facilities;
- Fact Census 2021 (INE), which integrates socioeconomic variables derived from the 2021 Census, including demographic indicators, income, employment, and household characteristics;
- Fact Rent Evolution, which records rental values over time, enables the analysis of temporal trends and spatial dynamics in the rental market.

3.9 DASHBOARD DESIGN

The dashboard design is grounded in four guiding principles that structure both visual organization and analytical use. First, an information hierarchy was adopted, moving from overview to detail and, finally, to explanation, allowing users to begin with a global view of the Rental Stress Index (RSI) and progressively explore its spatial variability and underlying drivers. Second, the design places strong emphasis on territorial centrality, using the parish (freguesia) as the core unit of analysis and navigation. All visualizations support direct territorial selection and immediate feedback, reinforcing the spatial interpretation of rental stress. Third, a clear conceptual separation is maintained between the RSI as a synthetic indicator and the explanatory dimensions that contextualize it. While the index provides an overall measure of rental stress, complementary structural and contextual indicators are presented to support interpretation and avoid overly simplistic readings.

The dashboard architecture supports a structured analytical flow organized into three levels. At the first level, monitoring and pattern detection are enabled through a choropleth map and summary KPIs, which allow the identification of parishes with higher levels of rental stress and the recognition of spatial patterns such as clusters, gradients, and intra-urban contrasts. The second level focuses on territorial exploration and comparison, using filters and selectors to compare parishes, examine rankings and distributions, and identify relative differences between areas, thereby strengthening diagnostic capacity. The third level supports interpretation and contextualization by linking the RSI to a set of explanatory variables, including income levels, rental market characteristics, tourism pressure, and socio-demographic vulnerability. Together, these complementary visualizations facilitate an integrated reading of the dynamics underlying rental stress across the urban territory.



4. RESULTS AND DISCUSSION

4.1 RENTAL STRESS OVERVIEW

The Rental Stress Overview dashboard (Figure 3) provides a synthetic yet spatially detailed representation of rental stress across Lisbon's parishes. The average Rental Stress Index (RSI) stands at 35.55, while the maximum value reaches 100, indicating substantial territorial variation. This dispersion confirms that rental stress is not uniformly distributed across the municipality, but rather concentrated in specific high-pressure areas.

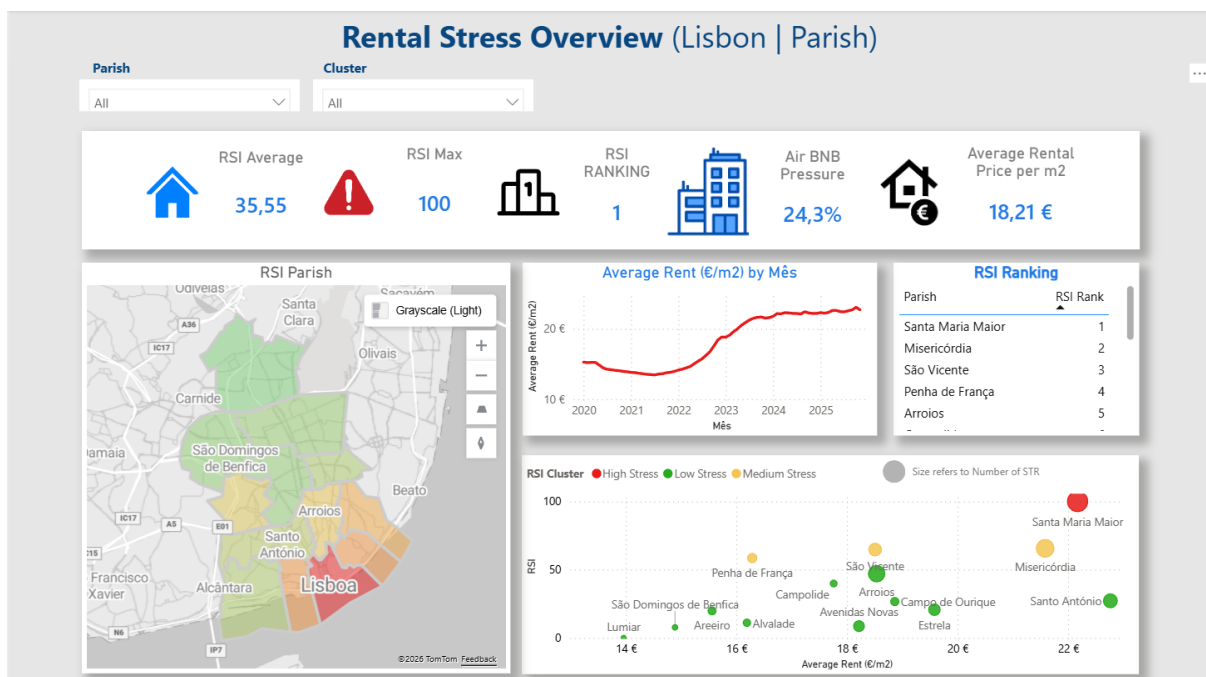


Figure 3 - Rental Stress Overview Page

The spatial visualization clearly highlights a cluster of high-stress parishes in the historical and central zones of the city. Santa Maria Maior ranks first in the RSI ranking, followed by Misericórdia and São Vicente. These parishes correspond to Lisbon's most consolidated urban core, characterized by high tourism intensity, dense built environments, and strong real estate demand. The map reveals a pronounced center-periphery gradient, where central parishes exhibit significantly higher stress levels compared to northern and western areas such as Lumiar or São Domingos de Benfica, which fall into the low-stress cluster.

The temporal evolution of average rental prices per square meter further contextualizes this pattern. Rental values show a clear upward trajectory from 2020 onwards, accelerating particularly after 2022. The current average rental price of €18.21 per m² reflects a structural increase rather than a temporary fluctuation. This sustained growth reinforces the interpretation of rental stress as a systemic and persistent urban phenomenon.



Another critical indicator displayed on this dashboard is Airbnb pressure, which stands at 24.3%. This figure suggests that nearly one quarter of the housing-related market activity in certain parishes is associated with short-term rental dynamics. The coexistence of high Airbnb pressure and elevated RSI levels in central parishes indicates a spatial overlap between tourism intensity and rental stress. Although causality cannot be inferred directly from this visualization alone, the territorial coincidence supports theoretical arguments linking short-term rental concentration with increased housing pressure and reduced long-term rental availability.

The bubble chart positioned at the bottom right of the dashboard further illustrates the relationship between average rent and RSI levels. Parishes classified as “High Stress” combine both elevated rental prices and high RSI values, with bubble size reflecting the volume of short-term rentals. Santa Maria Maior stands out as an extreme case, simultaneously exhibiting the highest RSI, among the highest rental prices, and a large STR footprint. In contrast, parishes such as Lumiar or São Domingos de Benfica present lower rent levels and correspondingly lower RSI values, reinforcing the central role of price dynamics in shaping stress intensity.

From a research perspective, these findings directly address Research Question 2 by revealing clear spatial patterns of intra-urban inequality. The RSI successfully distinguishes between high-pressure and low-pressure territories, validating its capacity to detect localized stress dynamics. Moreover, the dashboard demonstrates the analytical value of integrating spatial visualization with key performance indicators, thereby contributing to Research Question 3 regarding the usefulness of a Business Intelligence tool for policy monitoring.

4.2 RENTAL MARKET STRUCTURE

The Rental Market Structure page (Figure 4) provides a structural reading of Lisbon’s rental market, combining price levels, rental allocation, and contractual composition. The average rental price stands at €18.21 per m², yet clear territorial disparities emerge. Central parishes such as Santo António, Santa Maria Maior, and Misericórdia exhibit the highest rental values, confirming that price pressure is spatially concentrated in consolidated urban cores. Peripheral areas display comparatively lower rent levels, reinforcing the center–periphery gradient previously identified.

The allocation of housing stock to the rental sector further clarifies supply-side dynamics. With an average rental share of 42.27%, less than half of Lisbon’s classic dwellings are dedicated to long-term rental use. In high-demand central areas, a substantial rental share coexists with elevated prices, suggesting strong investor interest and intensified market competition. Conversely, parishes with lower rental allocations may face access constraints due to limited availability, which can indirectly contribute to price escalation. This pattern is consistent with scarcity-based explanations of housing market pressure (Lorga et al., 2022).

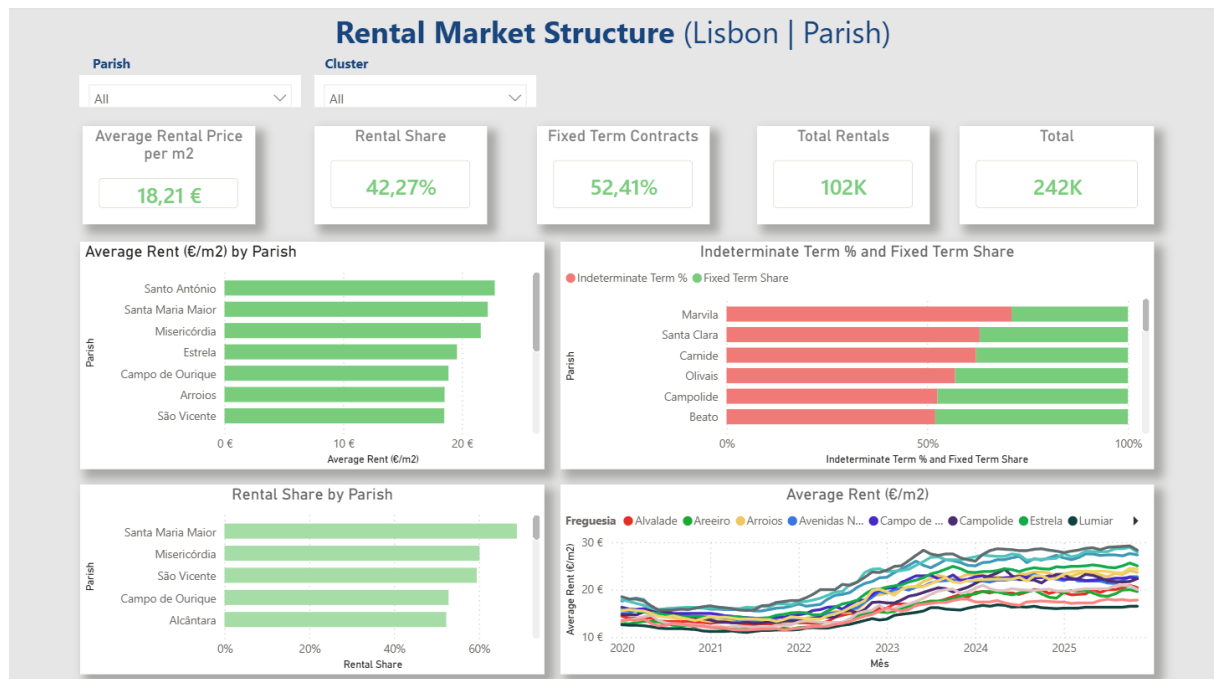


Figure 4 - Rental Market Structure Page

Contractual structure introduces an additional layer of vulnerability. Fixed-term contracts account for 52.41% of total agreements, exceeding indeterminate contracts. The higher prevalence of fixed-term arrangements in several high-demand parishes suggests greater contractual flexibility but also increased residential insecurity. As highlighted by Dancygier and Wiedemann (2025), contractual flexibilization can amplify housing stress by exposing tenants to more frequent renegotiations and potential rent increases. Therefore, rental stress emerges not solely from price intensity but also from institutional market characteristics.

The temporal evolution of rents confirms the structural dimension of these dynamics. Since 2020, rental prices have followed a sustained upward trajectory across most parishes, with acceleration after 2022. The widespread nature of this increase indicates systemic pressures rather than localized anomalies, aligning with broader processes of housing financialization and urban restructuring documented in Lisbon (Cocola-Gant & Gago, 2021).

4.3 TOURISM PRESSURE

The Tourism Pressure & Housing dashboard (Figure 5) highlights the scale and spatial distribution of short-term rental (STR) activity in Lisbon and its relationship with rental stress. Approximately 17,000 Airbnb units are registered, with an average Airbnb pressure of 17%, indicating a significant presence of short-term rental dynamics within the urban housing system.

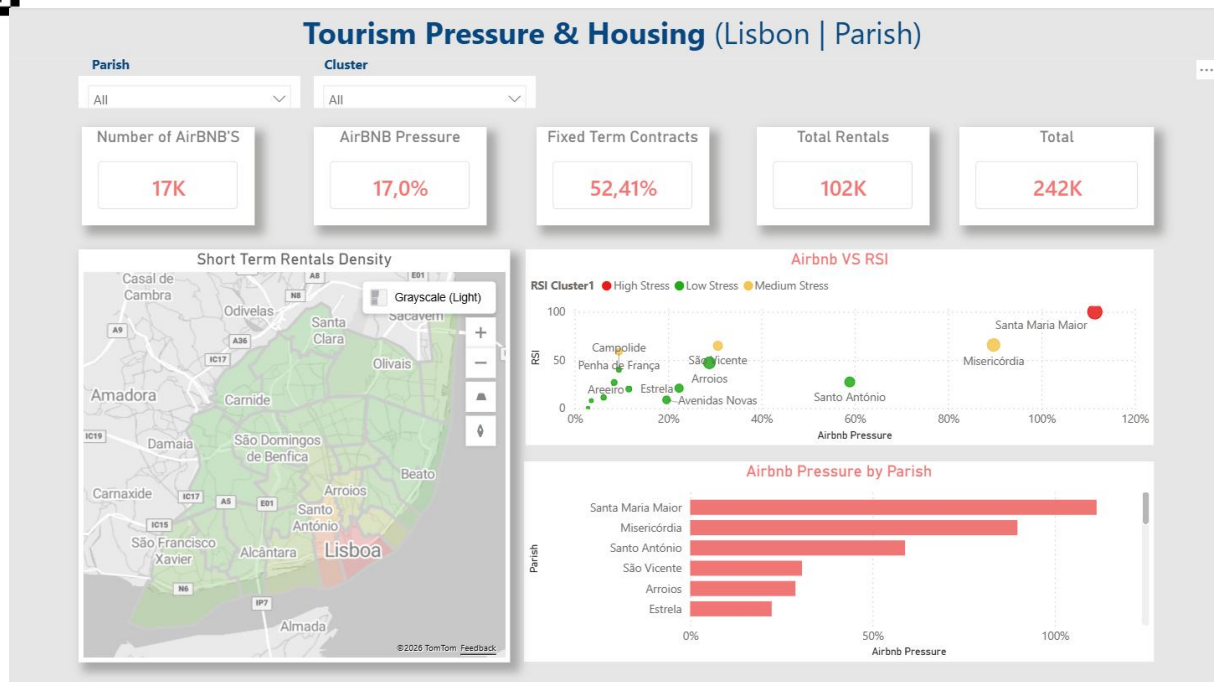


Figure 5 - Tourism Pressur & Housing Page

Spatially, Airbnb density is heavily concentrated in central parishes, particularly Santa Maria Maior and Misericórdia, followed by Santo António and São Vicente. These same parishes also record elevated RSI levels, suggesting a strong territorial overlap between tourism intensity and rental stress. The scatterplot relating Airbnb pressure to RSI further reinforces this association: parishes with higher STR pressure tend to display higher stress levels, with Santa Maria Maior emerging as an extreme case, combining a very high Airbnb presence with the highest RSI score.

This pattern is consistent with existing research arguing that short-term rental expansion can reduce effective long-term rental supply and intensify price pressure in high-demand urban cores (Lestegás et al., 2019; Lee & Kim, 2023). In Lisbon, where tourism has become structurally embedded in the urban economy, the conversion of residential units into STR listings may contribute to supply constraints and increased competition for housing. Although the dashboard does not establish causality, the spatial coincidence between STR concentration and rental stress aligns with quasi-experimental findings showing that STR restrictions can influence housing prices (Gonçalves et al., 2023).

Importantly, the interaction between tourism pressure and contractual structure further deepens vulnerability. With fixed-term contracts representing more than half of rental agreements, areas under strong tourism pressure may experience both price escalation and residential instability. This cumulative effect reflects broader processes of housing financialization and touristification documented in Southern European cities (Cocola-Gant & Gago, 2021).



4.4 SOCIOECONOMIC VULNERABILITY

The Socioeconomic Vulnerability dashboard (Figure 6) introduces demographic and income-related dimensions that contextualize rental stress across Lisbon. The unemployment rate stands at 4.05%, suggesting moderate overall labor market conditions. However, the aging index of 179.40 reveals a structurally aging population, indicating that in many parishes the number of residents aged 65 and over substantially exceeds the population under 15. This demographic imbalance has important implications for housing vulnerability, particularly in contexts where elderly residents rely on fixed or limited incomes.

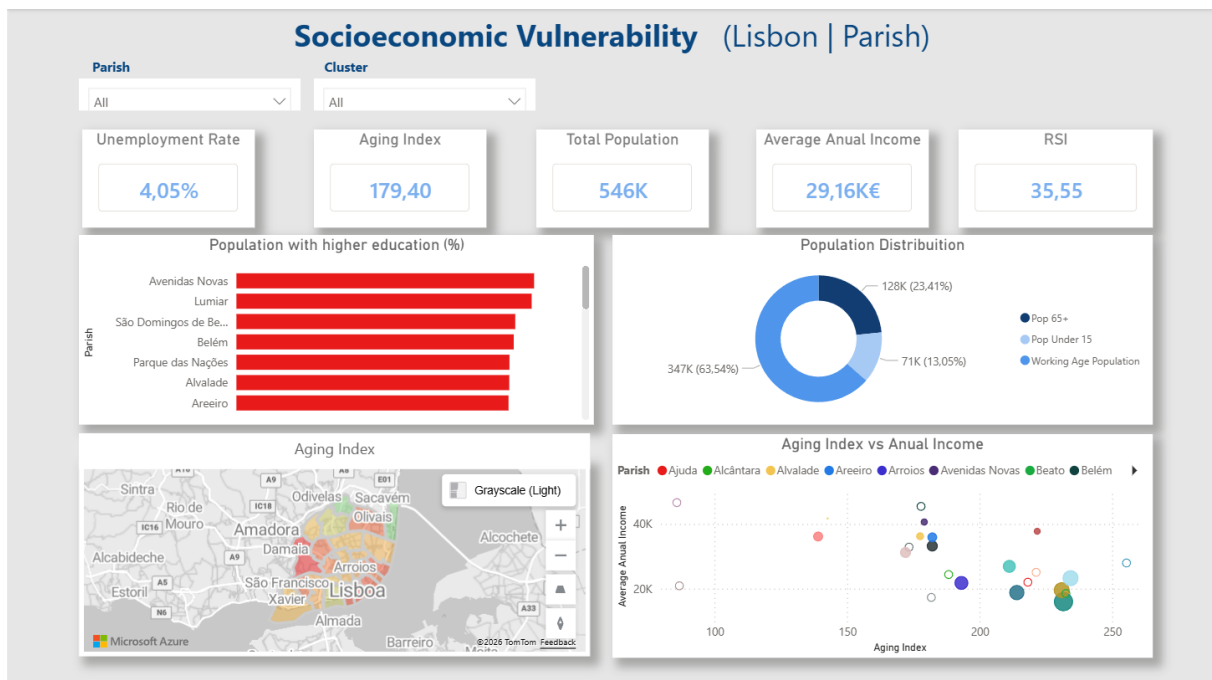


Figure 6 - Socioeconomic Vulnerability

The average annual income reaches €29,160, yet territorial disparities remain visible. The scatterplot relating the aging index to annual income highlights differentiated vulnerability profiles across parishes. Some areas combine high aging levels with relatively modest income, increasing exposure to rental pressure due to limited financial flexibility. Others exhibit higher income levels but still experience elevated RSI scores, suggesting that rental stress is not exclusively confined to low-income territories.

The population distribution further reinforces this interpretation. With approximately 63% of residents in working age and 23% aged 65 or over, the demographic structure becomes a critical factor in understanding housing resilience. As noted in housing affordability research, socioeconomic vulnerability interacts with market dynamics to shape stress exposure (Lee et al., 2022). Even moderate unemployment levels may amplify the rental burden where housing costs are rapidly increasing.



Spatially, parishes with higher aging indexes often overlap with areas experiencing moderate to high RSI values. This pattern suggests that demographic structure can intensify stress effects, particularly where housing markets are tight. The literature emphasizes that affordability challenges are multidimensional and sensitive to household composition and income stability (Biljanovska, 2023). The findings in Lisbon support this view, indicating that rental stress emerges from the intersection of price escalation and varying socioeconomic capacity to absorb costs.

4.5 URBAN ATTRACTIVENESS AND POINTS OF INTEREST

The Points of Interest (POI) Home Page dashboard (Figure 7) provides an overview of Lisbon's urban attractiveness by quantifying the distribution of commercial, cultural, educational, leisure, mobility, and safety infrastructures across parishes. On average, Lisbon records 2.94 POIs per 1,000 inhabitants, with a total of approximately 2,000 POIs identified. While the RSI average remains 34.49, the territorial distribution of amenities reveals important contextual differences that help explain variations in rental stress.

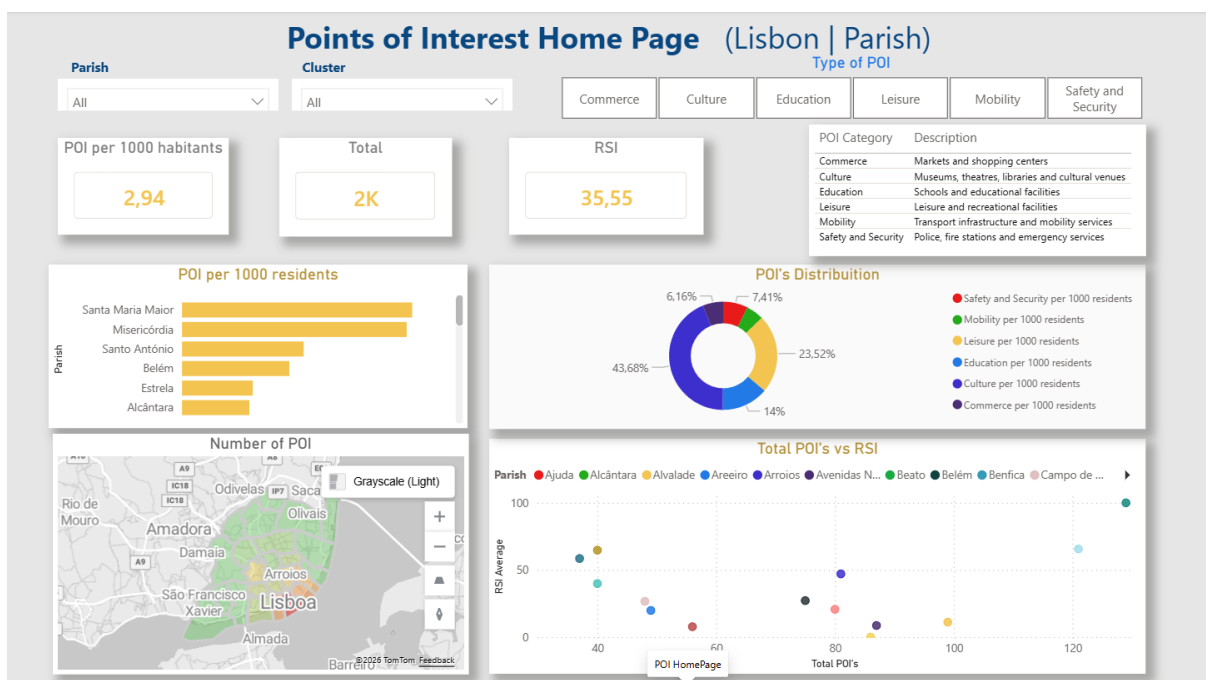


Figure 7 - Points of Interest Home Page Page

Central parishes such as Santa Maria Maior and Misericórdia display the highest concentration of POIs per 1,000 residents, reflecting their multifunctional character and strong urban centrality. The distribution chart indicates that commerce and culture account for a substantial share of total amenities, reinforcing their role as consolidated economic and tourism hubs. Peripheral parishes exhibit lower POI densities, suggesting weaker functional centrality and potentially lower demand pressure.



The scatterplot relating total POIs to RSI levels suggests a moderate positive association: parishes with a higher concentration of amenities tend to exhibit higher rental stress. This pattern is consistent with hedonic pricing theory, which argues that proximity to services, cultural venues, transport infrastructure, and commercial activity is capitalized into housing prices (Yang et al., 2020; Cheng et al., 2024). In highly attractive urban cores, the accumulation of amenities increases residential desirability, intensifying demand and, consequently, price pressure.

However, the relationship is not purely linear. Some parishes with moderate POI density present lower RSI levels, indicating that amenities alone do not determine stress intensity. Instead, urban attractiveness operates as a contextual amplifier when combined with high tourism intensity and constrained housing supply. This reinforces the multidimensional interpretation of rental stress, where structural market pressures interact with locational advantages.

4.6 INDICATORS OF RENTAL STRESS

The Indicators of Rental Stress dashboard (Figure 8) provides a decomposition of the Rental Stress Index (RSI), allowing the identification of the main drivers shaping stress intensity across Lisbon's parishes. The analysis confirms that rental price per square meter emerges as the primary differentiating indicator. With an average of €18.21/m², price exerts the strongest positive contribution to the RSI, reinforcing its central role in housing affordability dynamics.

The driver contribution analysis shows that rental price has the largest upward effect on the index, while average household income acts as a mitigating factor. This inverse relationship is consistent with affordability frameworks emphasizing the income–cost interaction as a core determinant of housing stress (Biljanovska, 2023). However, income alone does not offset structural price escalation in high-demand areas, which explains why several parishes maintain elevated RSI levels despite moderate income levels.

Additional contributing factors include short-term rental density, unemployment rate, aging index, and the share of fixed-term contracts. Short-term rental density increases stress by intensifying competition for housing supply, a mechanism documented in tourism-driven housing markets (Lee & Kim, 2023). The share of fixed-term contracts reflects contractual instability, which can amplify residential insecurity even when price levels are comparable (Dancygier & Wiedemann, 2025). Demographic aging further interacts with income constraints, particularly where older populations rely on fixed or limited financial resources.

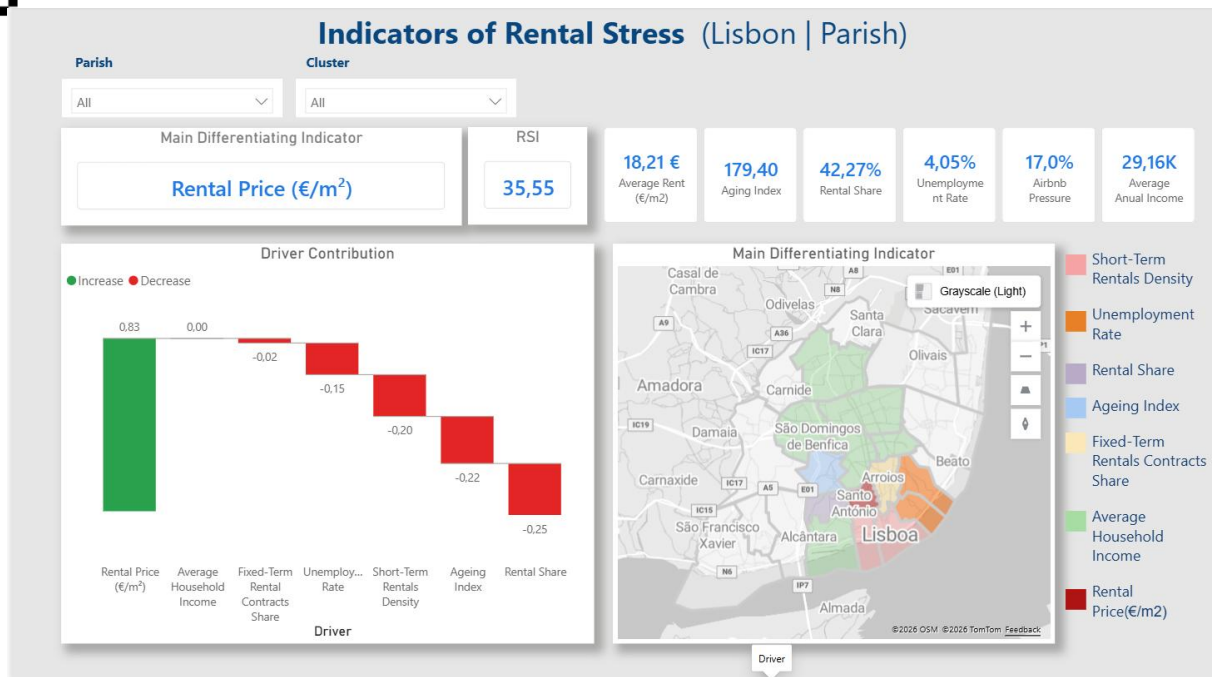


Figure 8 - Indicator os Rental Stress Page

Importantly, the decomposition demonstrates that rental stress is not driven by a single variable, but results from the cumulative interaction of economic, demographic, and structural market factors. While price remains the dominant force, the combined contribution of tourism pressure, contractual structure, and socioeconomic conditions reinforces the multidimensional validity of the RSI. This supports Research Question 1 by confirming that the composite index meaningfully captures the latent construct of rental stress.



5. CONCLUSIONS AND FUTURE RESEARCH

This study set out to develop a spatially explicit Rental Stress Index (RSI) capable of capturing the multidimensional nature of housing affordability pressures in Lisbon. The results confirm that rental stress is not a homogeneous phenomenon, but rather a spatially uneven and structurally embedded condition, concentrated in central parishes such as Santa Maria Maior, Misericórdia, and São Vicente. At the same time, peripheral areas exhibit comparatively lower levels of pressure. This center–periphery gradient reflects not only differences in price levels, but also deeper processes of urban restructuring, including tourism intensification, housing financialization, and the reconfiguration of residential space.

Beyond descriptive patterns, the findings reinforce a critical insight: rental stress cannot be reduced to a single economic variable, such as the rent-to-income ratio. Instead, it emerges from the interaction between housing market dynamics, socioeconomic vulnerability, institutional arrangements, and territorial characteristics. While rental price remains the dominant driver, the combined influence of short-term rentals, contractual instability, and demographic structure highlights the systemic nature of the phenomenon. In this sense, the RSI succeeds in operationalizing rental stress as a latent construct, rather than a direct observable measure, aligning with more advanced approaches in housing affordability research.

At the same time, the study reveals important epistemological and methodological tensions inherent in the construction of composite indices. The process of selecting, normalizing, and aggregating variables is not neutral, but rather shapes the very meaning of what is being measured. This becomes particularly evident in the treatment of Points of Interest (POIs).

The inclusion of POIs significantly alters the index's spatial configuration. This result is not merely a technical issue, but a conceptual turning point in the study. It exposes a fundamental tension between two distinct but often conflated dimensions: housing affordability stress, rooted in economic vulnerability; and urban attractiveness, driven by accessibility, amenities, and centrality.

By incorporating POIs into the RSI, the index risks drifting away from its original construct and becoming a proxy for urban desirability rather than housing stress. This highlights the sensitivity of composite indicators to variable selection and raises broader questions about construct validity in spatial analysis.

This distinction is analytically crucial. The RSI without POIs represents a more theoretically grounded and defensible measure, consistent with the literature on housing affordability and rental stress. In contrast, POIs function as contextual amplifiers, helping explain why certain areas experience higher demand and price pressure, but do not constitute stress in themselves. This separation allows the analysis to move beyond simplistic interpretations and instead engage with the trade-off between economic vulnerability and urban attractiveness.



From a critical perspective, this trade-off reflects a broader contradiction in contemporary cities: the very factors that make urban areas attractive—accessibility, cultural amenities, economic opportunities—also contribute to the exclusion of lower-income residents through rising housing costs. In Lisbon, this dynamic is particularly evident in central parishes, where high amenity density, strong tourism pressure, and elevated RSI values coexist. The findings, therefore, suggest that rental stress should be understood not only as a market outcome but also as a spatial manifestation of uneven urban development.

The study also highlights the role of tourism, particularly short-term rentals, as a structural component of this dynamic. While the analysis does not establish causality, the strong spatial overlap between Airbnb density and high RSI values supports existing arguments regarding the commodification of housing and the displacement effects of tourism-driven demand. Importantly, this interaction is further intensified by the prevalence of fixed-term rental contracts, which increase residential instability and reinforce vulnerability.

In this context, the RSI and its associated dashboard offer a valuable contribution not only as analytical tools, but also as instruments of critical urban diagnosis. By making spatial inequalities visible and quantifiable, they challenge the implicit neutrality of housing markets and provide a basis for more informed and potentially transformative policy interventions.

However, the study also acknowledges its limitations, which are not merely technical but also conceptual. The reliance on cross-sectional data limits the ability to capture dynamic processes such as displacement or gentrification over time. The use of factor analysis, while methodologically robust, introduces assumptions regarding dimensionality and weighting that may influence results. Moreover, as a composite index, the RSI inevitably simplifies complex realities, potentially obscuring intra-parish heterogeneity and lived experiences of housing stress.

More fundamentally, the RSI measures relative stress rather than absolute affordability thresholds, which raises questions about how results should be interpreted in policy contexts. A parish classified as “low stress” may still face significant affordability challenges when considered in absolute terms. This highlights the importance of complementing composite indices with other analytical approaches and qualitative insights.

Future research should therefore move in several directions. First, integrating longitudinal data would enable analysis of temporal dynamics and the identification of processes such as displacement and neighborhood transitions. Second, applying causal inference methods could help disentangle the relationships among tourism, housing supply, and price formation. Third, expanding the framework to a metropolitan or comparable scale would enhance its generalizability and enable cross-city benchmarking.

The challenge for policymakers is not only to reduce rental stress but to confront the underlying dynamics that produce it. Ensuring that Lisbon remains a vibrant and attractive city must not come at the cost of social exclusion. Instead, it requires a deliberate effort to balance



economic development, urban livability, and social equity, recognizing that housing is not merely a market commodity but a fundamental component of the right to the city.



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